

NIDS Session Report

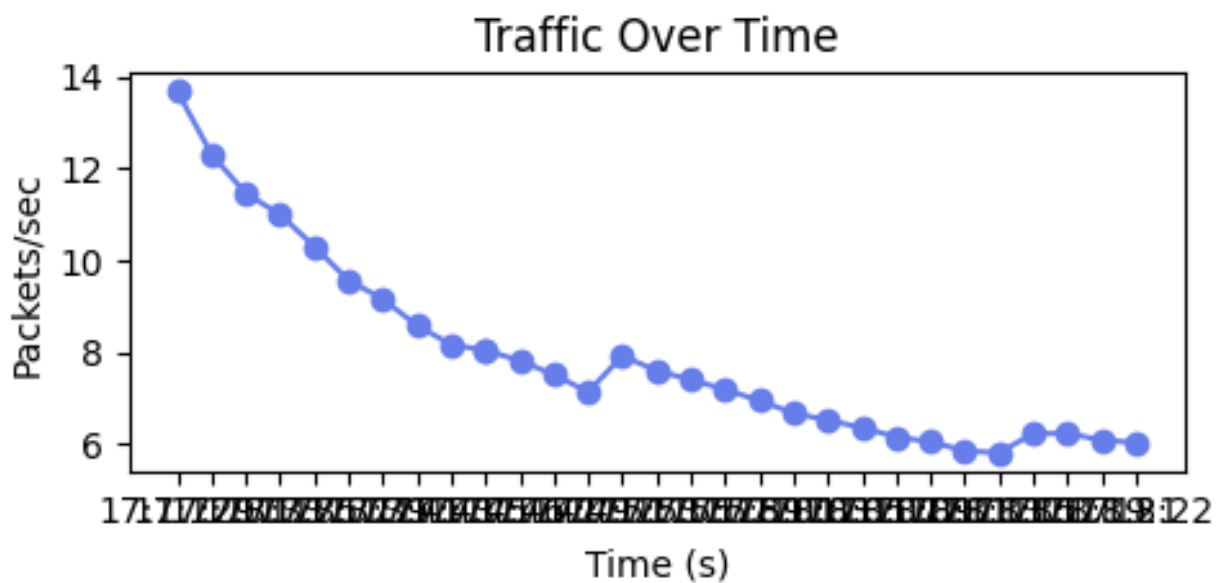
AI-Based Network Intrusion Detection System

This report summarizes your network activity and any suspicious events detected. It's designed to be easy to understand, even if you're not a network expert.

How to Read This Report

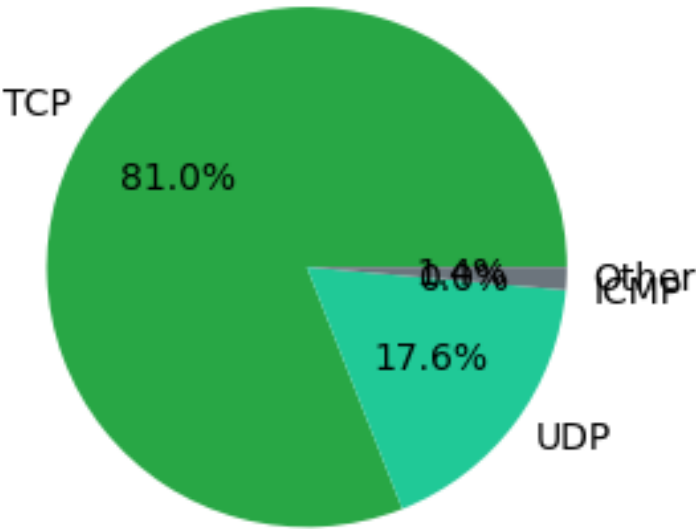
- Packets: Each piece of data sent over your network.
- Anomaly: Something unusual or unexpected in the network traffic.
- Alert: A warning about a specific suspicious activity.

Traffic Over Time



Protocol Distribution

Protocol Distribution



Session Summary

Start Time:	1750247231.738361
Stop Time:	2025-06-18T17:18:22.221851
Uptime:	70 seconds
Packets Processed:	426
Anomalies Detected:	13
Packet Rate:	6.04 packets/sec

Protocol Statistics

TCP PACKETS:	345
UDP PACKETS:	75
ICMP PACKETS:	0
OTHER PACKETS:	6

Security Alerts (6)

Type: Port Scan | Time: 2025-06-18T17:17:13.017563

Details: {'src_ip': '10.0.2.15', 'dst_ip': '10.0.2.3', 'dst_port': 53, 'protocol': 17}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: Ddos Indicator | Time: 2025-06-18T17:17:13.552376

Details: {'src_ip': '34.231.73.62', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: A large number of packets were sent to a single device, possibly overwhelming it. This is a common sign of a Distributed Denial of Service (DDoS) attack, which can take down services.

Type: MI Anomaly | Time: 2025-06-18T17:17:16.780808

Details: {'anomaly_score': -3.5723526270285233, 'features': {'packet_count': 100, 'total_bytes': 25127, 'avg_packet_length': np.float64(251.27), 'std_packet_length': np.float64(436.871373633018), 'min_packet_length': 54, 'max_packet_length': 1545, 'avg_payload_size': np.float64(195.99), 'std_payload_size': np.float64(437.4897140505133), 'total_payload_size': 19599, 'tcp_count': 96, 'udp_count': 4, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.96, 'udp_ratio': 0.04, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 15, 'unique_dst_ports': 19, 'total_ports_accessed': 19, 'common_port_ratio': 0.42, 'high_port_ratio': 0.58, 'avg_dst_port': np.float64(31889.23), 'std_dst_port': np.float64(26787.6498054066), 'min_dst_port': 53, 'max_dst_port': 60775, 'unique_connections': 23, 'total_connection_attempts': 100, 'avg_packets_per_connection': 4.3478260869565215, 'connection_diversity': 0.23, 'max_packets_per_connection': 26, 'unique_src_ips': 14, 'unique_dst_ips': 11, 'src_ip_diversity': 0.14, 'dst_ip_diversity': 0.11, 'private_src_ip_ratio': 0.14285714285714285, 'private_dst_ip_ratio': 0.18181818181818182, 'packet_rate': 25.695473489020852, 'inter_arrival_time_mean': np.float64(0.039310464955339526), 'inter_arrival_time_std': np.float64(0.0996004583529076), 'burst_count': 67, 'burst_ratio': 0.6767676767676768, 'payload_size_mean': np.float64(195.99), 'payload_size_std': np.float64(437.4897140505133), 'zero_payload_ratio': 0.25, 'large_payload_ratio': 0.1, 'payload_size_entropy': np.float64(1.1241277142044424), 'syn_count': 24, 'fin_count': 6, 'rst_count': 0, 'psh_count': 27, 'ack_count': 86, 'urg_count': 0, 'syn_ratio': 0.25, 'fin_ratio': 0.0625, 'rst_ratio': 0.0, 'psh_ratio': 0.28125, 'ack_ratio': 0.8958333333333334, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T17:17:16.780795', 'packet_count': 100, 'protocol_distribution': {'tcp': 96, 'udp': 4, 'icmp': 0}, 'connection_info': {'unique_connections': 23, 'unique_src_ips': 14, 'unique_dst_ips': 11}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

Type: Port Scan | Time: 2025-06-18T17:18:14.727670

Details: {'src_ip': '10.0.2.15', 'dst_ip': '10.0.2.3', 'dst_port': 53, 'protocol': 17}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: MI Anomaly | Time: 2025-06-18T17:18:16.951008

Details: {'anomaly_score': -3.5683181783125097, 'features': {'packet_count': 100, 'total_bytes': 29324, 'avg_packet_length': np.float64(293.24), 'std_packet_length': np.float64(445.4720893613875), 'min_packet_length': 54, 'max_packet_length': 1292, 'avg_payload_size': np.float64(242.08), 'std_payload_size': np.float64(450.16578457275045), 'total_payload_size': 24208, 'tcp_count': 56, 'udp_count': 44, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.56, 'udp_ratio': 0.44, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 15, 'unique_dst_ports': 33, 'total_ports_accessed': 33, 'common_port_ratio': 0.32, 'high_port_ratio': 0.68, 'avg_dst_port': np.float64(37039.65), 'std_dst_port': np.float64(25122.495866603305), 'min_dst_port': 53, 'max_dst_port': 55102, 'unique_connections': 31, 'total_connection_attempts': 100,

'avg_packets_per_connection': 3.225806451612903, 'connection_diversity': 0.31, 'max_packets_per_connection': 14, 'unique_src_ips': 19, 'unique_dst_ips': 14, 'src_ip_diversity': 0.19, 'dst_ip_diversity': 0.14, 'private_src_ip_ratio': 0.10526315789473684, 'private_dst_ip_ratio': 0.14285714285714285, 'packet_rate': 4.038040280643291, 'inter_arrival_time_mean': np.float64(0.2501463432504673), 'inter_arrival_time_std': np.float64(0.7147732318209291), 'burst_count': 54, 'burst_ratio': 0.5454545454545454, 'payload_size_mean': np.float64(242.08), 'payload_size_std': np.float64(450.16578457275045), 'zero_payload_ratio': 0.09, 'large_payload_ratio': 0.15, 'payload_size_entropy': np.float64(1.2018488151946873), 'syn_count': 45, 'fin_count': 0, 'rst_count': 4, 'psh_count': 1, 'ack_count': 48, 'urg_count': 0, 'syn_ratio': 0.8035714285714286, 'fin_ratio': 0.0, 'rst_ratio': 0.07142857142857142, 'psh_ratio': 0.017857142857142856, 'ack_ratio': 0.8571428571428571, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T17:18:16.950995', 'packet_count': 100, 'protocol_distribution': {'tcp': 56, 'udp': 44, 'icmp': 0}, 'connection_info': {'unique_connections': 31, 'unique_src_ips': 19, 'unique_dst_ips': 14}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

Type: Ddos Indicator | Time: 2025-06-18T17:18:17.111163

Details: {'src_ip': '3.234.165.166', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: A large number of packets were sent to a single device, possibly overwhelming it. This is a common sign of a Distributed Denial of Service (DDoS) attack, which can take down services.

What Should I Do?

- If you see many alerts: Consider checking your devices for unusual activity or malware.
- If you see DDoS or Port Scan alerts: Make sure your firewall is active and up-to-date.
- If unsure: Share this report with your IT support or a cybersecurity professional.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
Port Scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.
DDoS Indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
Unusual Protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ML Anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.